

Announcements

► Any questions/issues?

► Schedule:

Today (Follow-up): Discussion Q2 (absorption lines);
Discussion Q3 (**now a bonus question**)

Today (Follow-up): Boiling Lab — what did we observe and what does it mean?

Today (Follow-up): Where are we on T_{Earth} ?

Today (New): Why is there evaporation below boiling?

Tomorrow (Lab): Evaporation of water and clouds!

Thursday (Discussion): Final calculation of Earth's equilibrium temperature.

Next Week: No Lab; Starting paleo perturbations of temperature.

Today's Reading Quiz (3 min)

On your own sheet of paper. . .

Say a few words about each: *atmospheric pressure*,
humidity of air, *latent heat*.

Turn in to Blackboard as scanned pdf (or image is fine).

Question 2 from Discussion

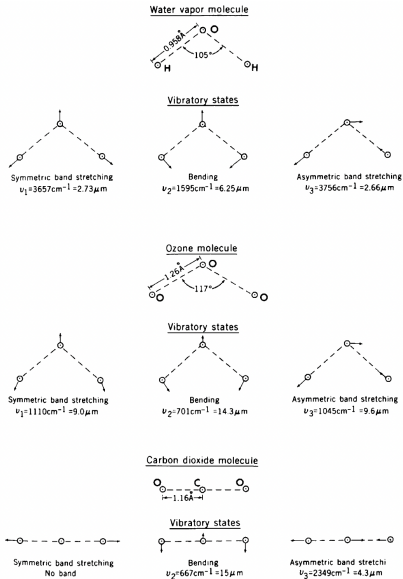
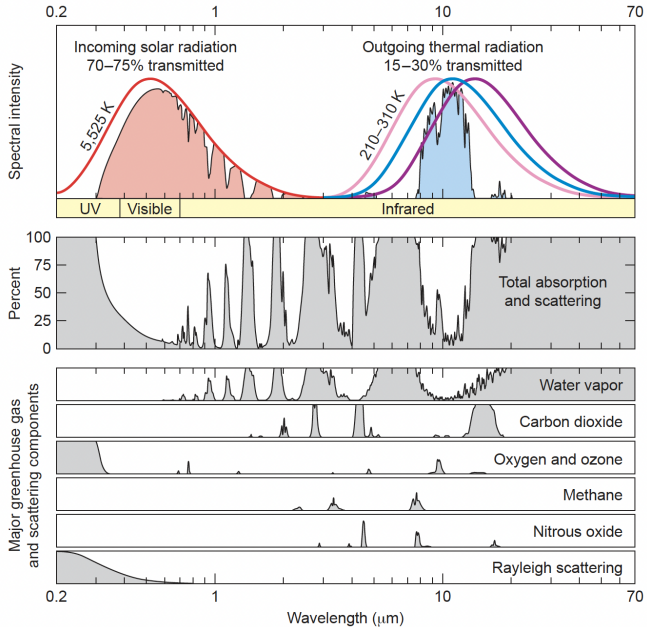


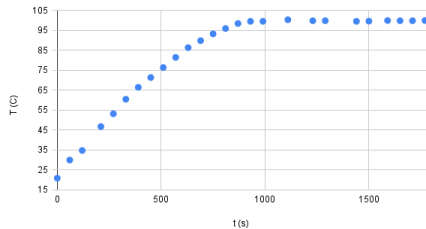
FIGURE 6.5. The structures of the water vapor, ozone, and carbon dioxide molecules and their vibratory states (from Houghton, 1985).

Question 3 from Discussion

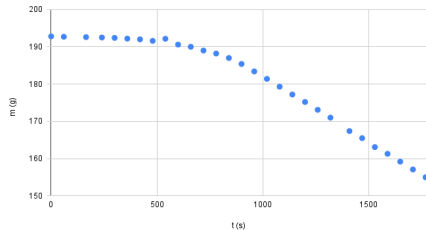


Boiling Lab: Data

T (C) vs. t (s)



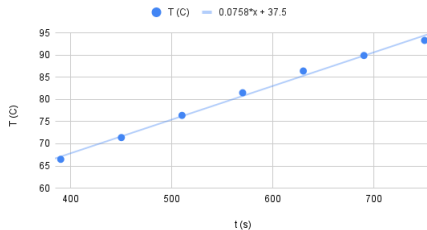
m (g) vs. t (s)



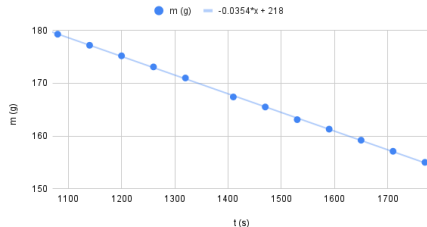
Boiling Lab: Analysis

Let's calculate the *latent heat of vaporization*, L_v , knowing that the specific heat capacity of water is 4.2 J/gK .

T (C) vs. t (s)

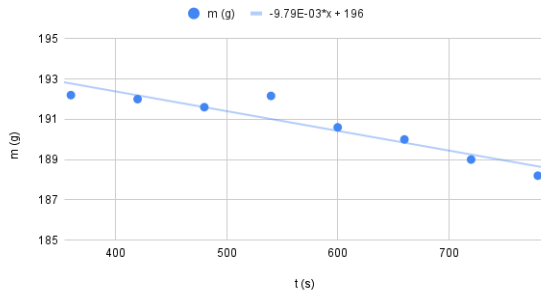


m (g) vs. t (s)

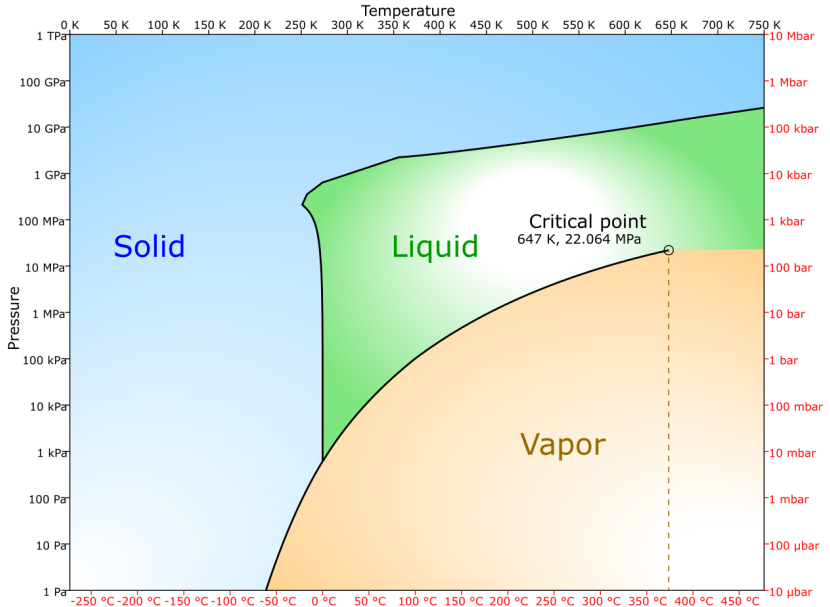


Boiling Lab: Analysis (2nd Attempt)

m (g) vs. t (s)



Phase Transitions in (Pure) Water



Water Evaporation in Air

Consider this figure from the *Mathez* textbook reading today:

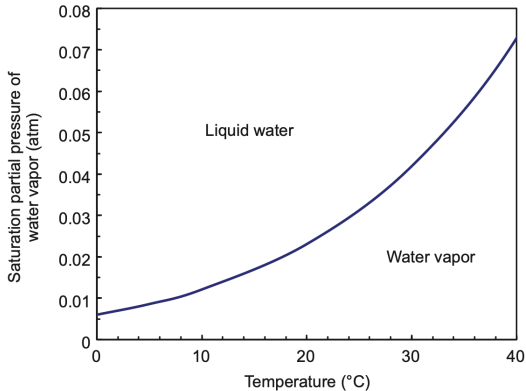


FIGURE 1.2 The water vapor pressure in air in equilibrium with liquid water as a function of temperature at 1 atmosphere total pressure

The curve was calculated from the Clausius-Clapeyron equation.

Why are the values of the pressure so different along the line of transition from liquid to vapor?

Water Evaporation in Air

Answer: Because evaporation in air uses the so-called **partial pressure** of water within the whole air:

$$p_{\text{partial, H}_2\text{O}} = \frac{\text{moles}_{\text{H}_2\text{O}}}{\text{moles}_{\text{air}}} \times p_{\text{atm}}$$

Where $p_{\text{atm}} = 101 \text{ kPa} = 1 \text{ atm}$ is the full atmospheric pressure. Typically the mole fraction above is a few percent. Let's try with 2% ... what is the new boiling point?

Calculating the equilibrium T_{Earth}

Recall our steps to calculate the equilibrium temperature of Earth thus far:

- ▶ Planet with no atmosphere, receiving solar radiation from the sun at $S/4(1 - \alpha)$
- ▶ Planet receiving that solar radiation, with an atmosphere.

Now (in Discussion this week) we will make one final adjustment:

- ▶ Planet with an atmosphere, plus transport of water vapor above the greenhouse gas layer.